Info 511 Final Project

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Why Health Insurance?

- As of 2023, 92% of Americans had health insurance (public or private) (Keisler-Starkey, K. & Bunch, L. N.)
- Average health insurance premium for an american family was $25,572 in 2024. An increase of 24% since 2019 (KFF).
- Summing to a cost of $318.4 billion dollars a year for Americans (Health Insurance - worldwide: Statista market forecast).

Is there a practical way for individuals to reduce how much their insurance company spends on them, and thereby reduce how much the individual spends on insurance?

Medical Insurance Cost Prediction Dataset

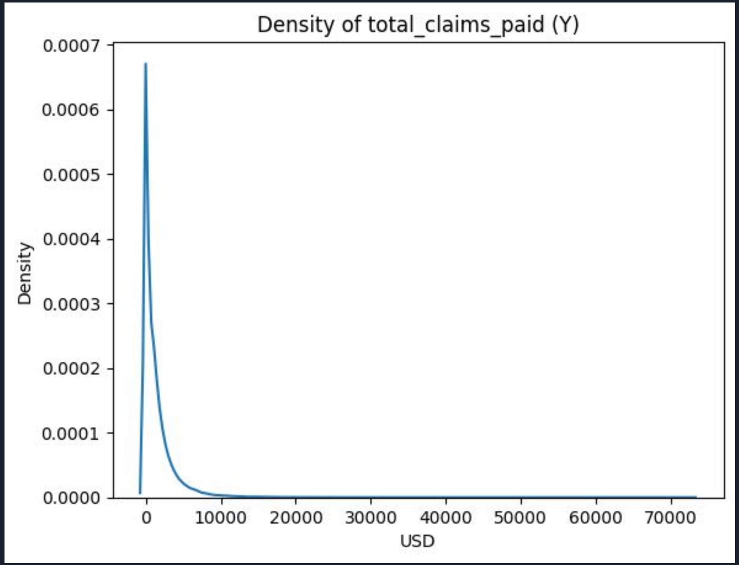
- Originates from Kaggle. Author is Mohan Krishna Thalla
- Dataset has information on 100,000 individuals such as demographics, health history, medical insurance expenditures, and lifestyle
- Data covers the globe and the year range of 2014-2024
- “The dataset was compiled from publicly available health surveys, insurance research studies, and anonymized online healthcare data” (Thalla)

Exploratory Data Analysis

Missing Value Analysis

Variable Name	P-value
Total_claims_paid (Y)	0.569 (t-test)
BMI	0.893 (t-test)
Income	0.734 (t-test)
Alcohol_freq	0.470 (Chi-sq test)
Employment_status	0.749 (Chi-sq test)
Urban_rural	0.079 (Chi-sq test)

Distribution of Target Variable



$$Y_{totalClaimsPaid} = \beta_0 + \beta_1 X_{income} + \beta_2 X_{bmi} + \beta_3 X_{smokerFormer} + \beta_4 X_{smokerCurrent} + \beta_5 X_{alcoholWeekly} + \beta_6 X_{alcoholDaily} + \beta_7 X_{suburban} + \beta_8 X_{rural} + \beta_9 X_{unemployed} + \beta_{10} X_{retired} + \epsilon_i$$

Model, Variables, and Description

Variable Name	Description
Income	Annual Income (USD)
BMI	Body Mass Index
Smoker_Former	Used to smoke
Smoker_Current	Currently smokes
Alcohol_Weekly	Drinks weekly
Alcohol_Daily	Drinks daily
Suburban	Lives in a suburban area
Rural	Lives in a rural area
Unemployed	Doesn’t have a job currently
Retired	No longer working

Models and methods

1. Linear model - No transformation
    a. Violated assumptions of normality and linearity

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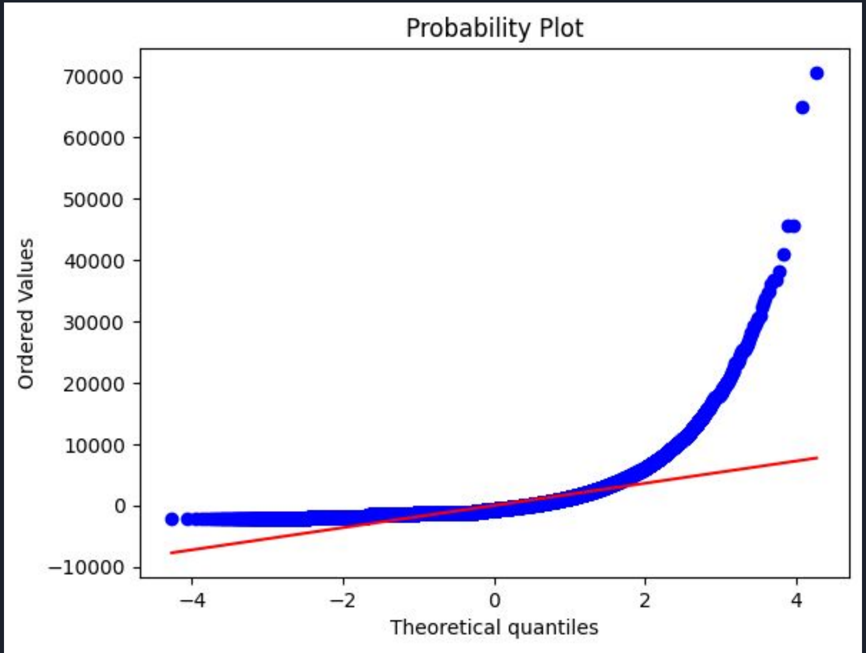
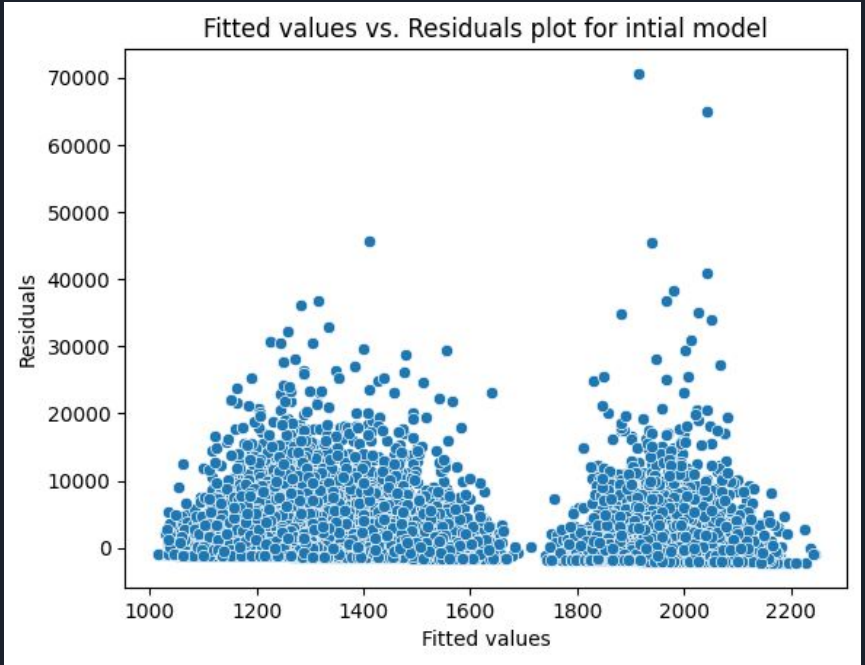
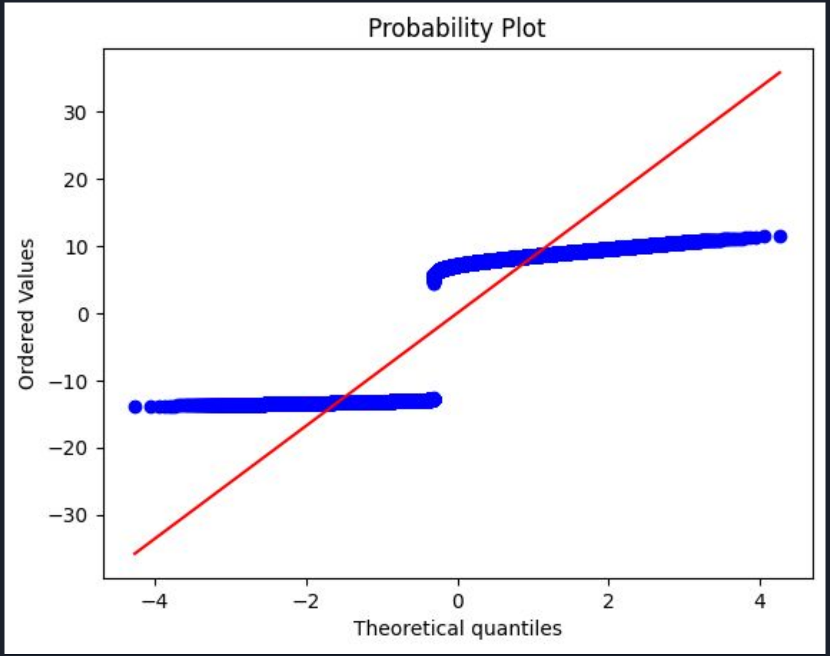


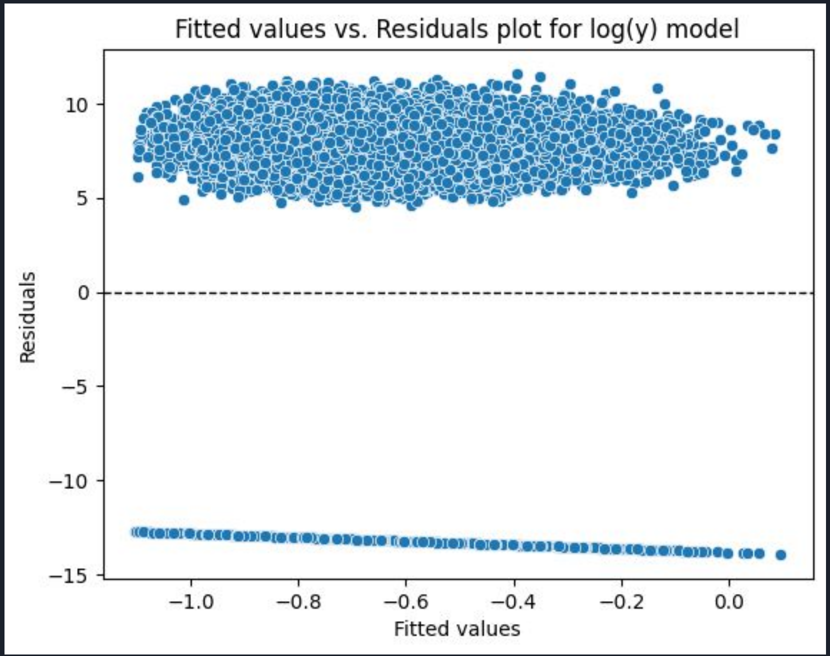


Models and methods

1. Linear model - No transformation
    a. Violated assumptions of normality and linearity
2. Linear model - Log transformation on total_claims_paid (Y)
    a. Violated assumption of normality, linearity, and homoscedasticity
    b. *Strange* behavior obvious from qq-plot and fitted vs. residual plot
        i. Two clear groups formed! One for no claims and one for at least one claim paid

Linear model - Log transformation on total_claims_paid (Y)





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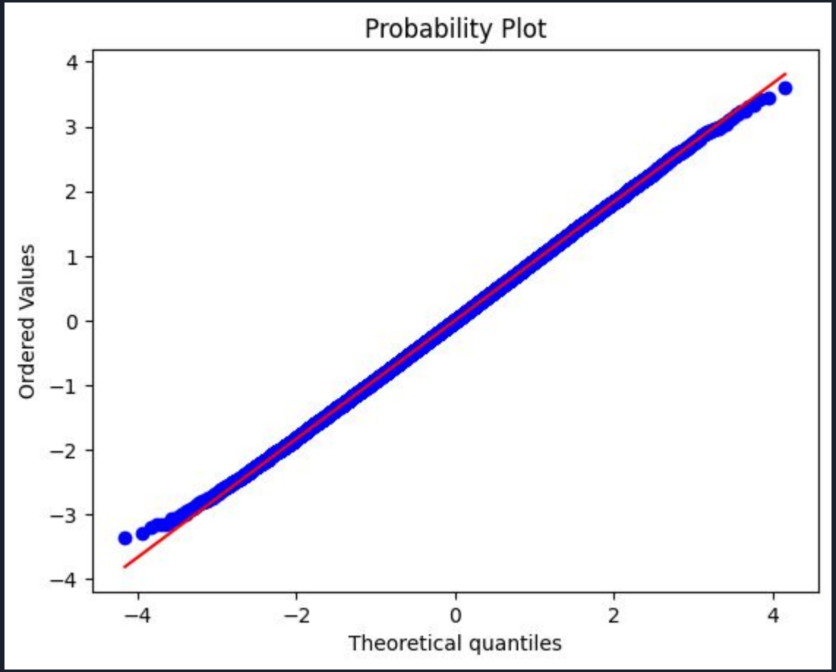
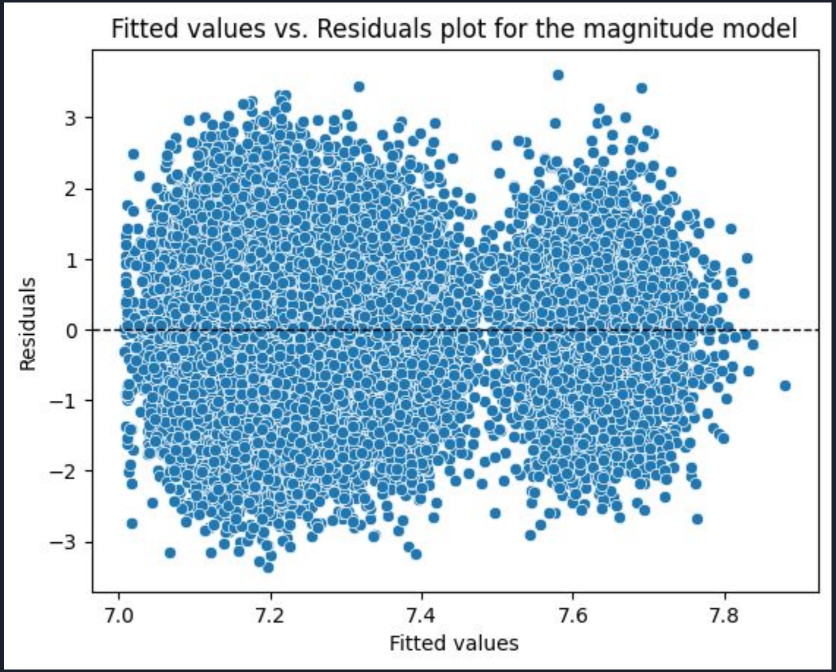
4. Final linear model - Log transformation on (Y) and only on individuals where at least $1 was spent
    a. All linear regression assumptions met
    b. Low predictive power

"Magnitude Model"





“Magnitude Model”

$$\log \hat{Y}_{totalClaims} = 6.88 - 0.00000003X_{income} + 0.011X_{bmi} + 0.144X_{smokeFormer} - 0.001X_{alcoholWeekly} +$$

$$0.022X_{alcoholDaily} + 0.006X_{suburban} + 0.018X_{rural} + 0.021X_{unemployed} - 0.003X_{retired}$$

Conclusion

- At an R^2 value of 0.029, our model doesn't explain enough variation to responsibly predict medical insurance claims paid
- However, we still have several variables that have coefficient estimates, which are significant.

95% Bonferroni Adjusted Family Confidence Intervals

Variable Name	Adjusted Confidence Interval (USD per year)
BMI	($1.008, $1.013)
Current_smoker	($1.521, $1.626)
Former_smoker	($1.042, $1.089)

Possible follow-ups

- Poisson regression on claims made in the year
- Linear regression on total medical costs (insurance + individual)
- Logistic regression including lifestyle choices AND prior medical history

References

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